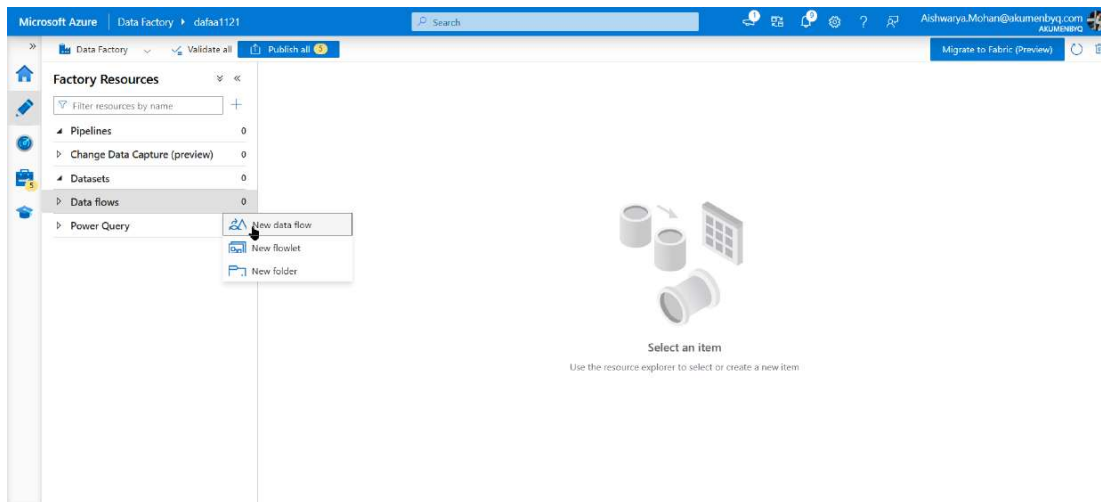


Azure Data Factory – Data Flow Activity

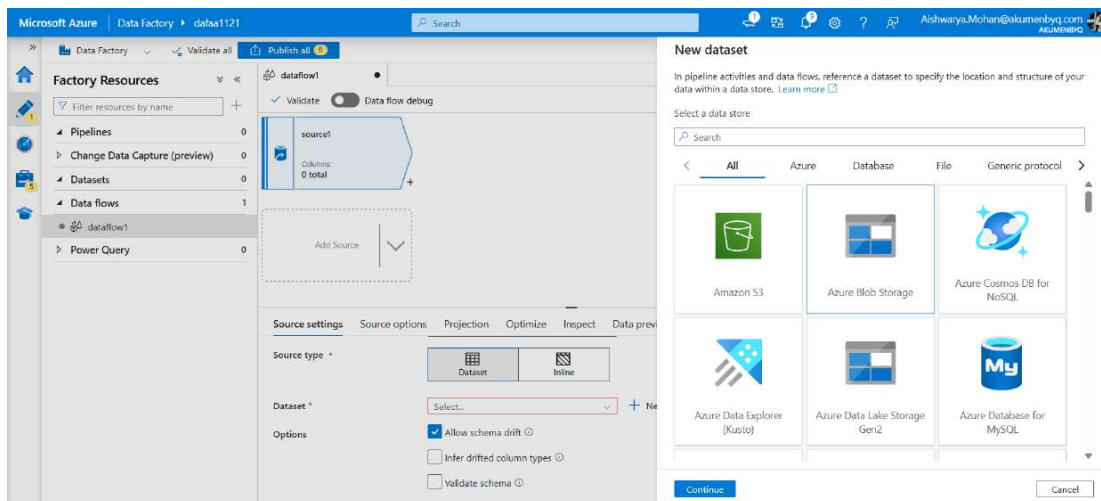
Source → Derived Column → Select → Sink

① Create Data Flow & Add Source

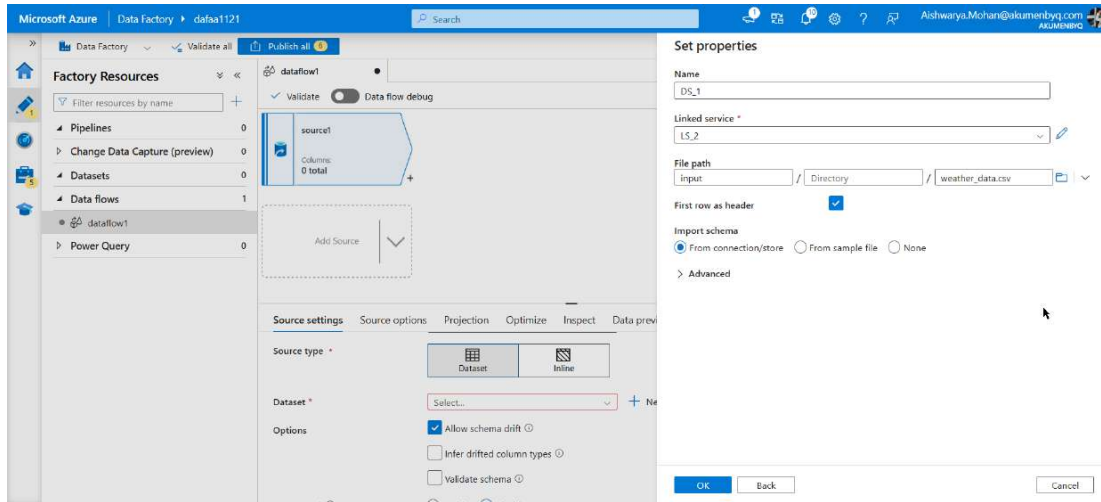
New data flow → Source type = Dataset → Azure Blob Storage



New data flow created → dataflow1



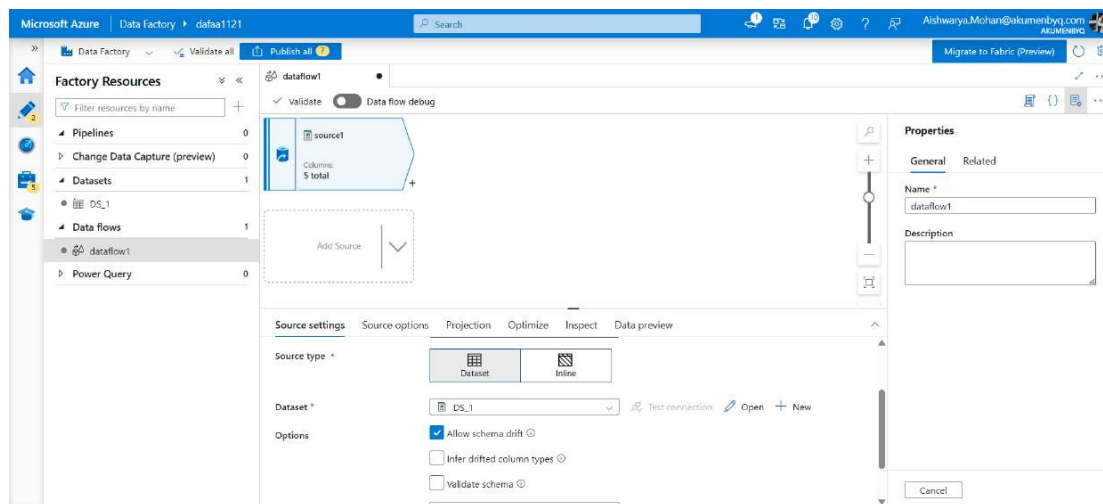
New dataset → Azure Blob Storage selected



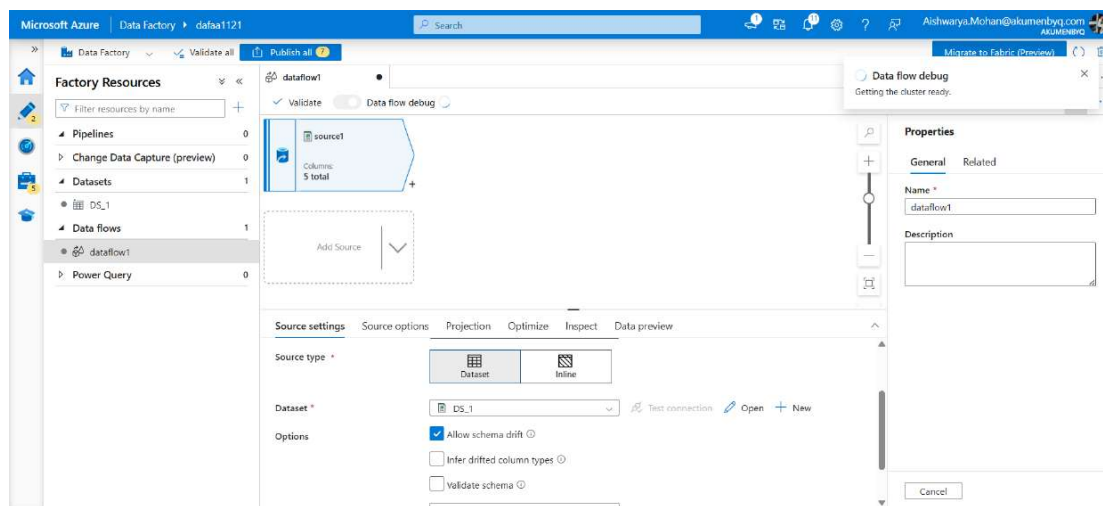
Set properties → DS_1 linked to weather_data.csv

② Import Schema & Enable Debug

source1 connected to DS_1 → Data flow debug turned on



source1 imports schema → 5 columns total



Data flow debug → getting the cluster ready

Microsoft Azure | Data Factory | data1121

Search

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Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - DS_1
- Data flows 1
 - dataflow1
- Power Query 0

dataflow1

Validate Data flow debug Debug Settings

source1

Columns: 5 total

Add Source

Source settings Source options Projection Optimize Inspect Data preview

Number of rows: INSERT 5 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 5

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	date	city	temperature_c	humidity_pct	condition
+	2026-08-19	New York	22.5	60	Sunny
+	2026-08-20	London	18.3	75	Cloudy
+	2026-08-21	Tokyo	25.1	68	Rainy
+	2026-08-22	Sydney	19.8	55	Clear
+	2026-08-23	Mumbai	30.2	80	Humid

Properties

General Related

Name *

dataflow1

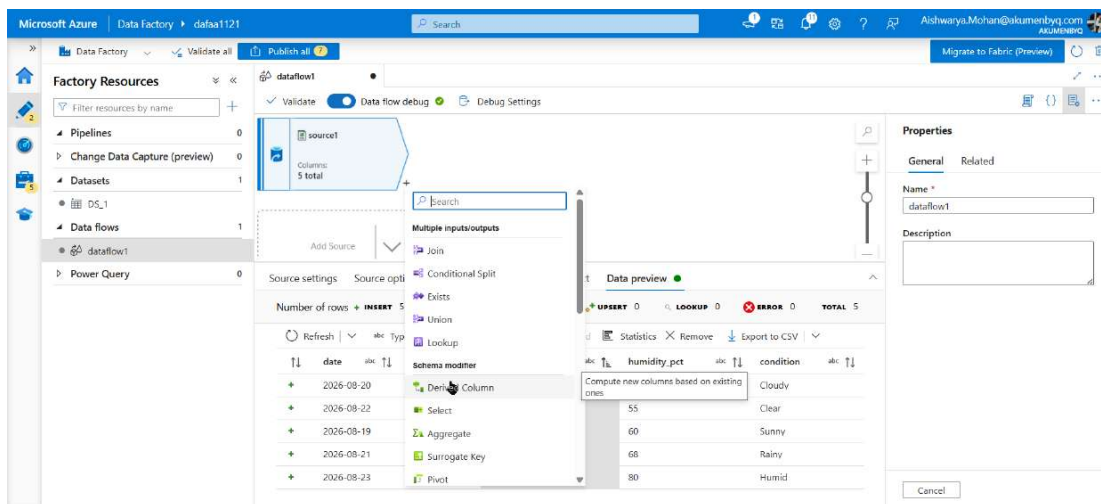
Description

Cancel

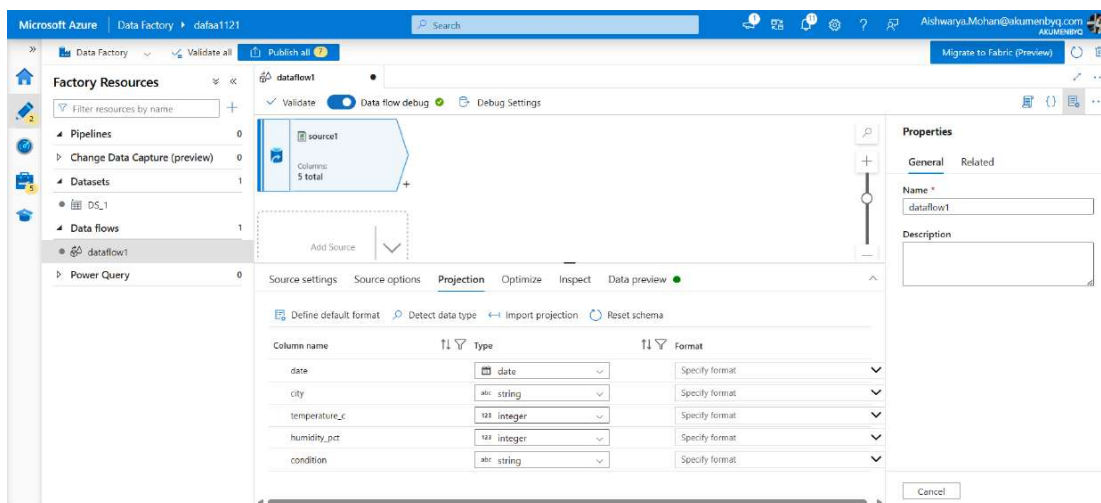
Data preview → 5 rows of weather data

③ Add Derived Column Transformation

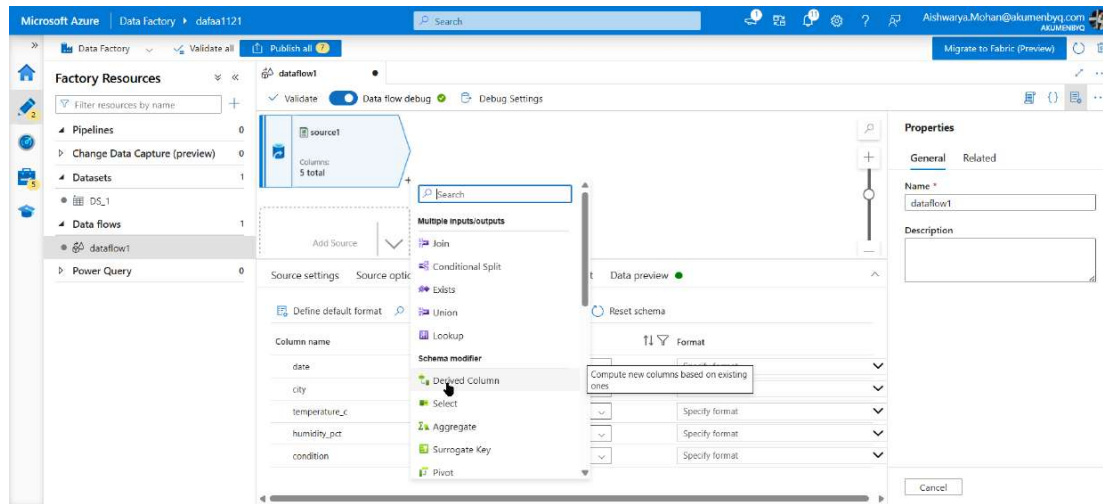
Schema modifier → Derived Column added after source1



Schema modifier menu → Derived Column selected



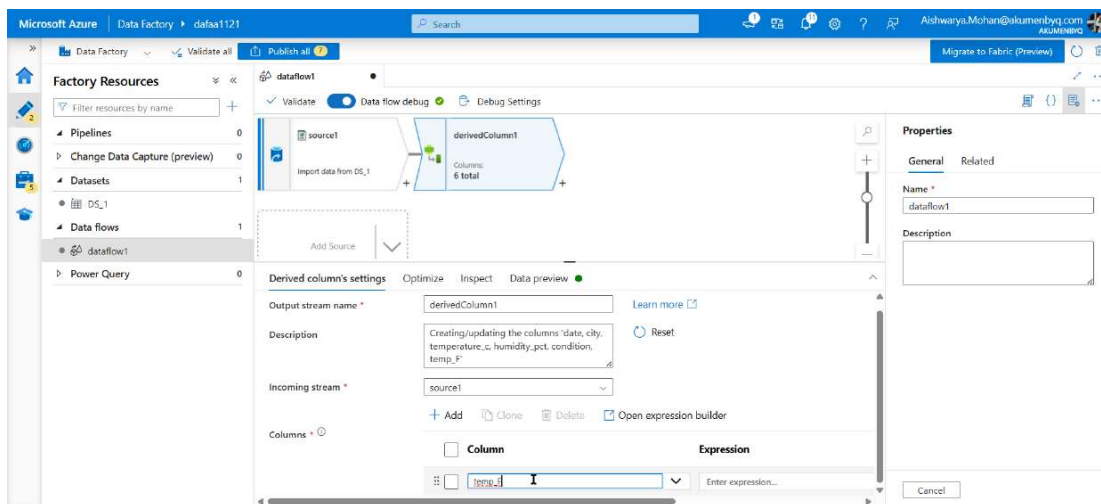
Projection → source columns: date, city, temperature_c, humidity_pct, condition



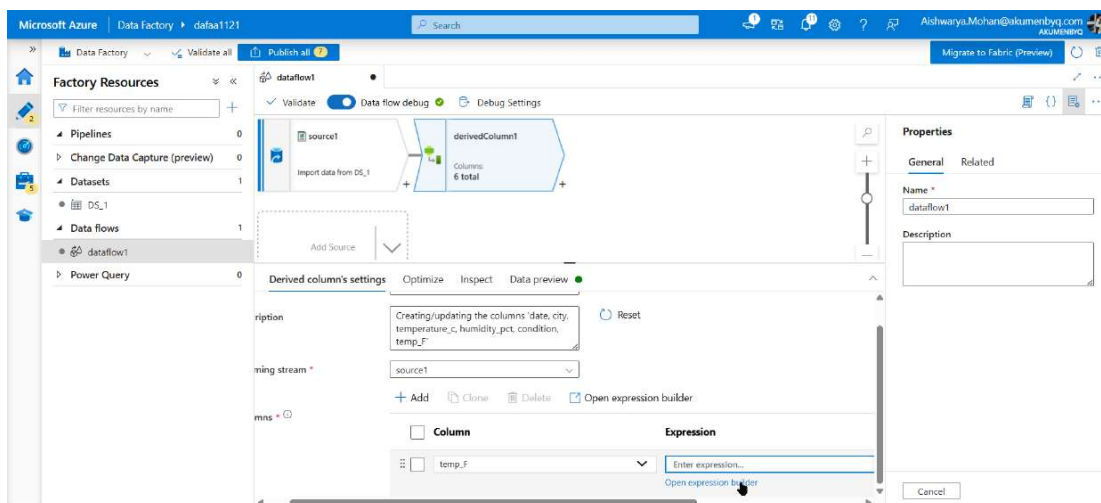
derivedColumn1 added → 6 total columns

④ Build the Expression

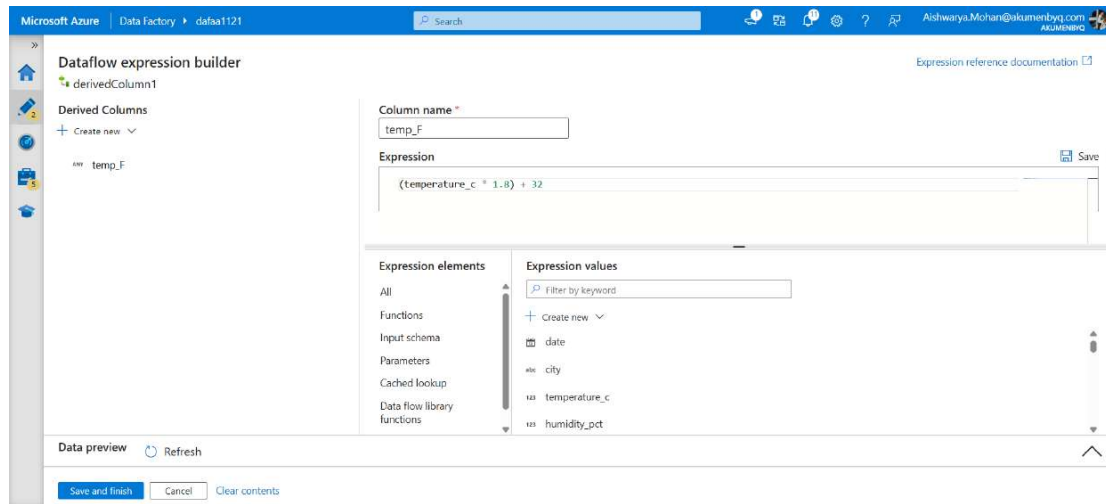
New column temp_F = (temperature_c * 1.8) + 32



New column name entered → temp_F



Open expression builder



*Expression builder → $(temperature_c * 1.8) + 32$*

⑤ Derived Column Result

Expression saved → temp_F visible in data preview

Microsoft Azure | Data Factory | dataa1121

Factory Resources

- Pipelines
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

dataflow1

Derived column's settings

Output stream name: derivedColumn1

Description: Creating/updating the columns: 'date, city, temperature_c, humidity_pct, condition, temp_F'

Input stream: source1

Columns:

Column	Expression
temp_F	(temperature_c * 1.8) + 32

Properties

General

Name: dataflow1

Description:

Expression saved → derivedColumn1 settings

Microsoft Azure | Data Factory | dataa1121

Factory Resources

- Pipelines
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

dataflow1

Data preview

Number of rows: INSERT 5, UPDATE 0, DELETE 0, UPSERT 0, LOGUP 0, ERROR 0, TOTAL 5

date	city	temperature_c	humidity_pct	condition	temp_F
2026-08-19	New York	22	60	Sunny	71.6
2026-08-20	London	18	75	Cloudy	64.4
2026-08-21	Tokyo	25	68	Rainy	77.0
2026-08-22	Sydney	19	55	Clear	66.2
2026-08-23	Mumbai	30	80	Humid	86.0

Properties

General

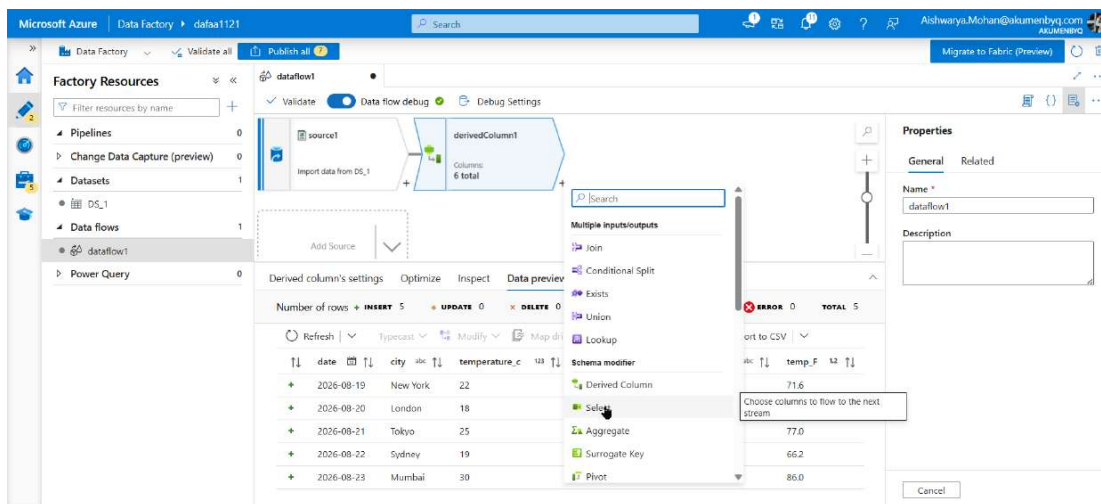
Name: dataflow1

Description:

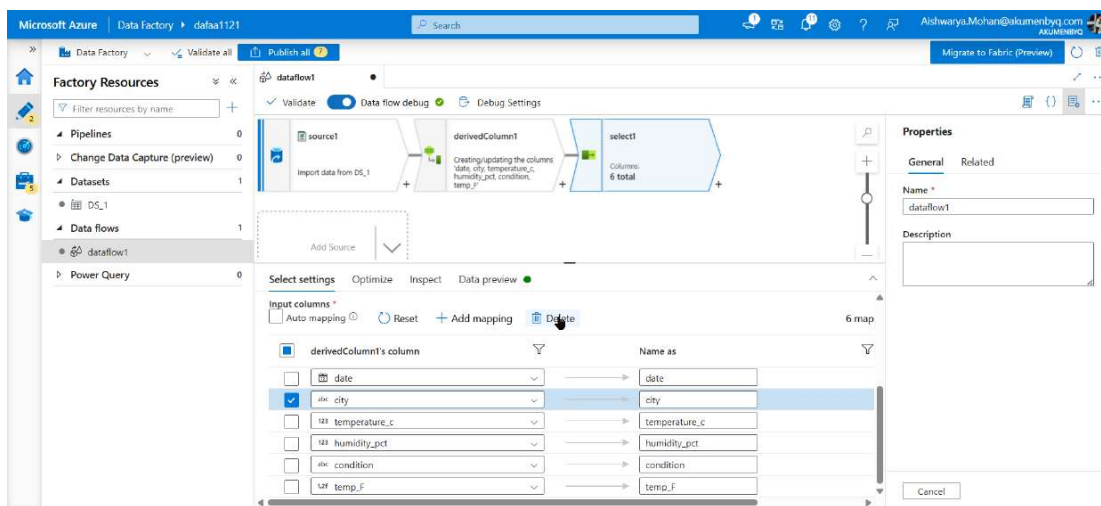
Data preview → temp_F column added

⑥ Add Select Transformation

Schema modifier → Select added → city column removed



Schema modifier menu → Select added



Select settings → city column unchecked / deleted

Microsoft Azure | Data Factory | data1121

Search

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Factory Resources

- Pipelines
- Change Data Capture (preview)
- Datasets
- DS_1
- Data flows
- dataflow1
- Power Query

dataflow1

Validate Data flow debug Debug Settings

source1 Import data from DS_1

derivedColumn1 Creating/updating the columns state, city, temperature_c, humidity_pct, condition, temp_f

select1 Columns: 5 total

Properties

General Related

Name * dataflow1

Description

Select settings Optimize Inspect Data preview

Number of rows INSERT 5 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 5

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

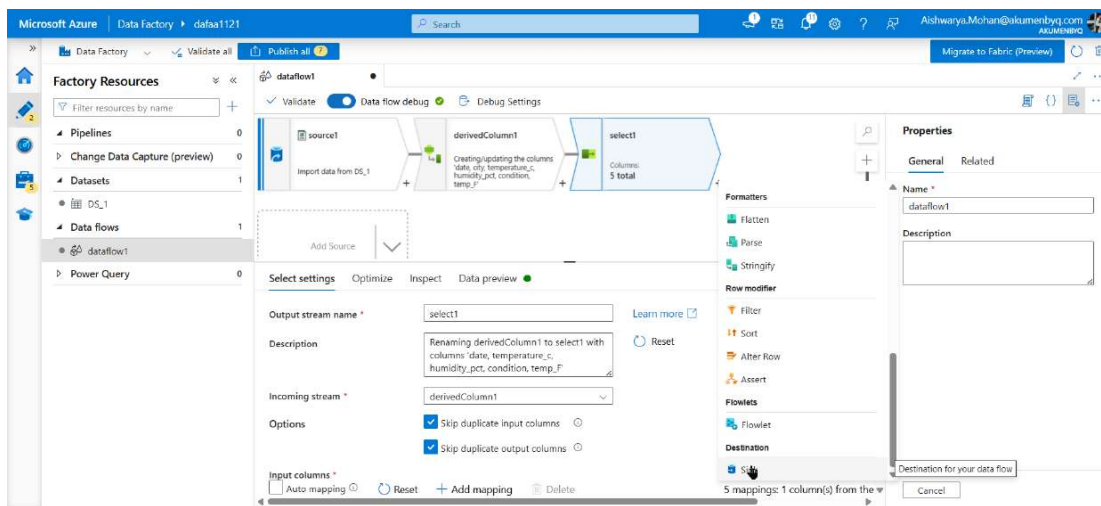
data	temperature_c	humidity_pct	condition	temp_F
2025-08-19	22	60	Sunny	71.6
2025-08-20	18	75	Cloudy	64.4
2025-08-21	25	68	Rainy	77.0
2025-08-22	19	55	Clear	66.2
2025-08-23	30	80	Humid	86.0

Cancel

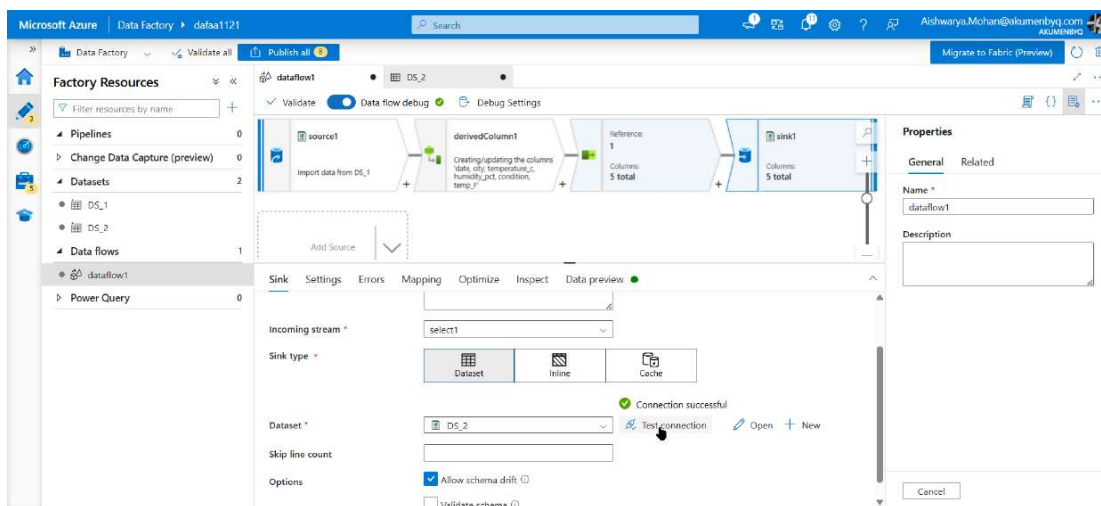
Data preview → city dropped, temp_F retained

⑦ Add Sink & Configure Output

Destination → Sink added → DS_2 (output container)



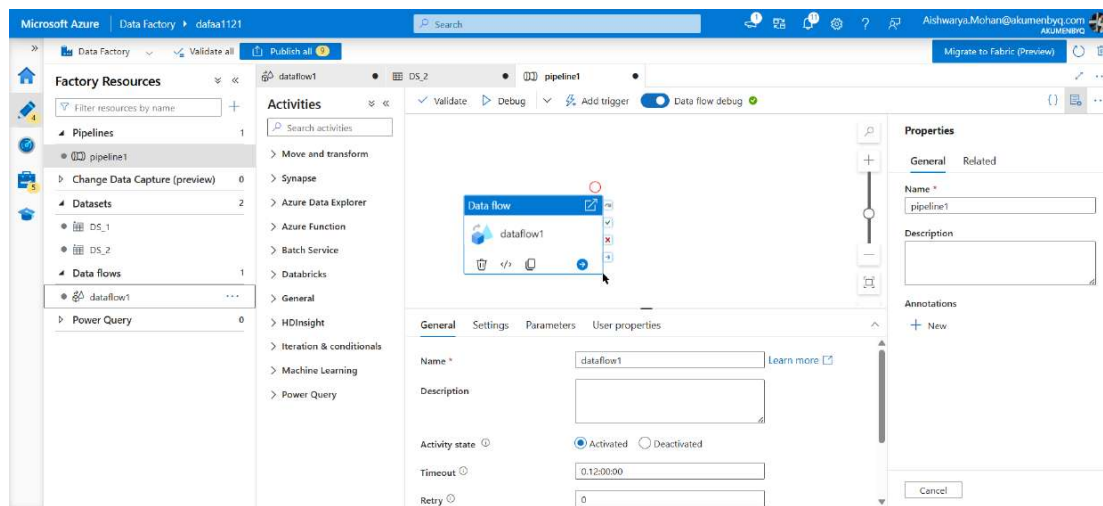
Destination menu → Sink added



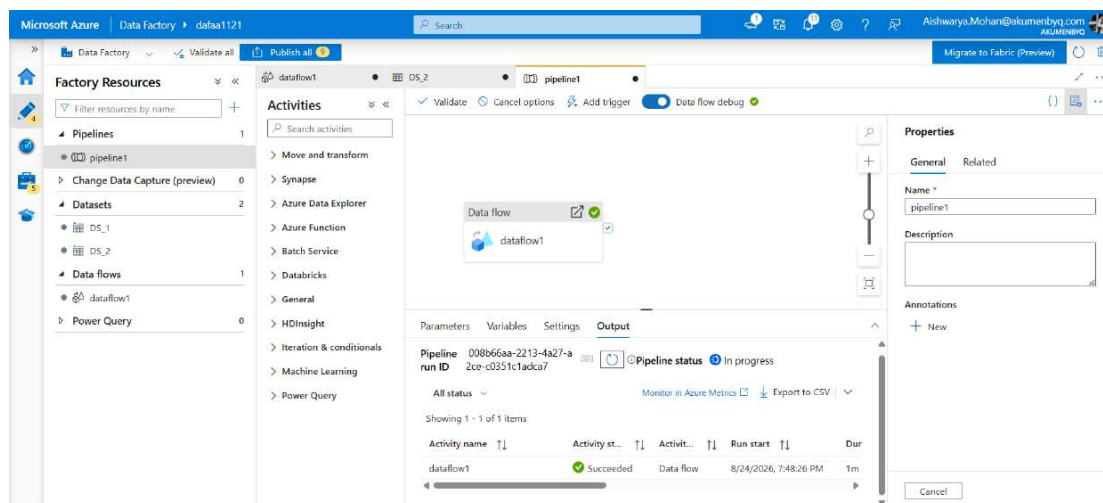
Sink settings → Dataset DS_2 → Test connection successful

⑧ Run the Pipeline

pipeline1 → Data flow activity → Debug run



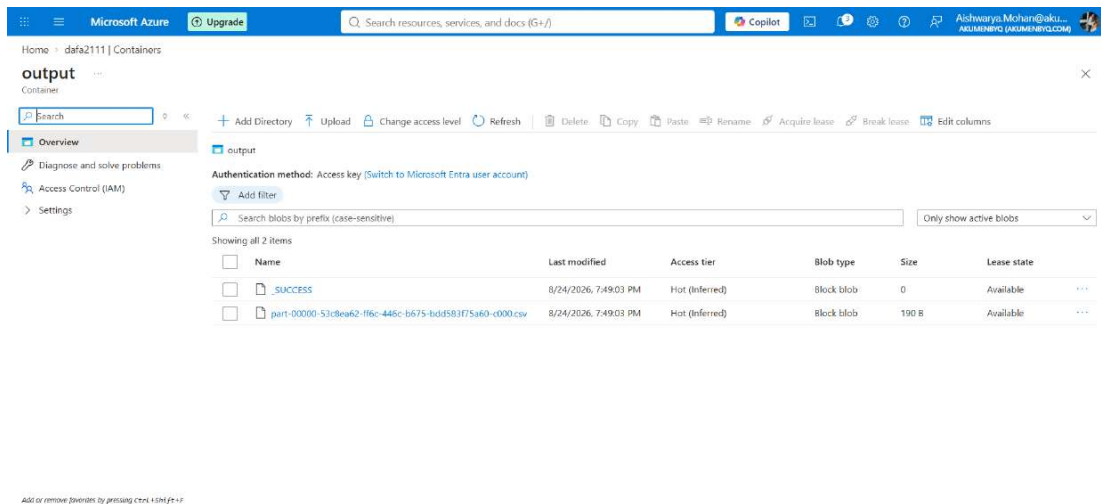
pipeline1 created → Data flow activity added



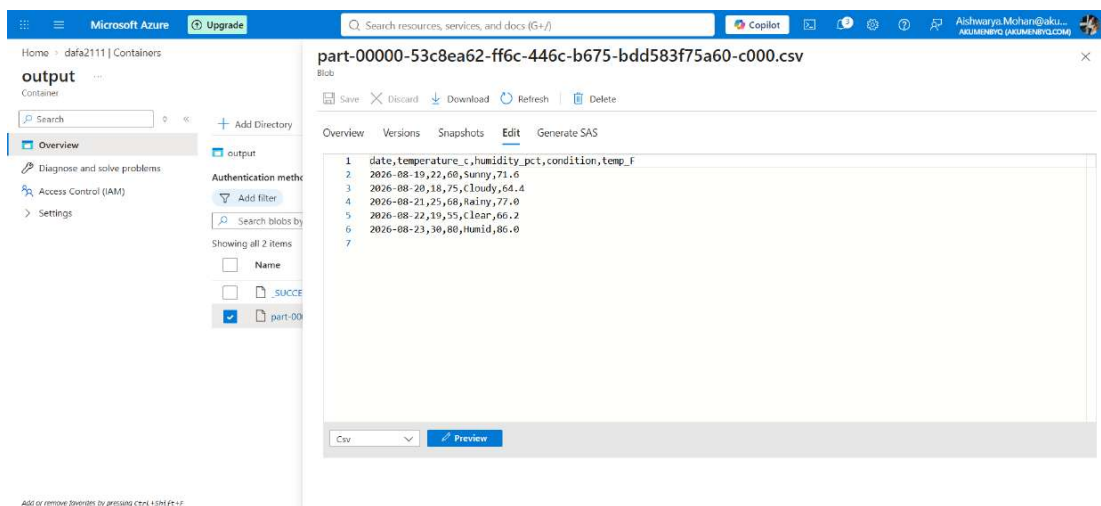
Pipeline debug run → dataflow1 succeeded

⑨ Verify Output in Blob Storage

output container → CSV file with temp_F column



Output container → SUCCESS file + part-00000 csv



CSV content → final data with temp_F column

Logic:

Source (weather_data.csv) → Derived Column (temp_F) → Select (drop city) → Sink (DS_2 / output container)